Intelligent Computing
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COMMENTARY

A Commentary on “Inferring Neural Activity before Plasticity as a Foundation for Learning beyond Backpropagation”

Xingsi Dong[1,2,3], Xiangyuan Peng[1,2,3], and Si Wu[1,2,3*]

[1]School of Psychology and Cognitive Sciences, Peking University, Beijing, China. [2]IDG/McGovern Institute for Brain Research, Peking University, Beijing, China. [3]Academy for Advanced Interdisciplinary Studies, Peking University, Beijing, China.

*Address correspondence to: siwu@pku.edu.cn

Citation: Dong X, Peng X, Wu S. A Commentary on “Inferring Neural Activity before Plasticity as a Foundation for Learning beyond Backpropagation”. *Intell. Comput.* 2025;4:Article 0244. https://doi.org/10.34133/icomputing.0244

Submitted 28 September 2025
Revised 5 November 2025
Accepted 6 November 2025
Published 4 December 2025

The recent publication “Inferring neural activity before plasticity as a foundation for learning beyond backpropagation” presents a novel and compelling learning mechanism designated “prospective configuration (PC)”. As this mechanism is realized within the framework of predictive coding networks (PCNs), this commentary adopts the term “PCN” for consistency. The research positions this PCN framework as a formidable and biologically plausible alternative to backpropagation, the prevailing paradigm in modern deep learning. The objectives of this commentary are to contextualize the paper’s contributions within the extensive history of energy-based models, to conduct a rigorous analysis of the critical advantages and inherent trade-offs of the PCN framework relative to backpropagation, and to deliberate on the architectural constraints and prospective research avenues that this work illuminates.

The EBM Lineage and Training Paradigms

The learning algorithm realized through predictive coding networks (PCNs) [1,2] is fundamentally an energy-based model (EBM), one whose training follows an expectation–maximization (EM)-like procedure: neural activity inference corresponds to the expectation step (E step), converging to a low-energy state, while synaptic plasticity corresponds to the maximization step (M step), adjusting weights to make this state more probable. This lineage traces back to Boltzmann machines (BMs) [3], where energy takes a quadratic form ($x^T Wx$). PCNs adopt a related but distinct formulation, defining energy as the squared error between activity and prediction $\left((y-Wx)^2\right)$, thereby shifting from a generic quadratic function to predictive error. A central principle across BMs, their hierarchical variants, and PCNs is “local energy”, enabling neuron-level dynamics and local learning rules without requiring globally broadcasted errors.

From a training perspective, 2 main paradigms exist: the strong clamp [3], used in BMs and generalized in contrastive Hebbian learning [4], where visible layers are tightly constrained, and the weak clamp, exemplified by equilibrium propagation [5], which allows freer evolution of outputs. Early formulations of PCNs [2] including prospective configuration (PC) [6] employed a strong-clamp regime, whereas subsequent work [7] demonstrated that PCNs can also operate under weak-clamp conditions, conceptually aligning them with equilibrium propagation. PC also bears close conceptual ties to target propagation (TP) [8], another influential local credit-assignment mechanism. TP explicitly computes local targets—the neural activities that would yield a desired output—and propagates these targets backward through the network. PC shows that during relaxation, the activities of a PCN tend to converge toward these local targets, particularly when only the output neurons are clamped. In this regime, hidden layers progressively evolve toward the local targets defined by TP, and layers closer to the output converge earlier.

The intellectual evolution of PCNs is noteworthy. They were initially found to be trainable via local Hebbian rules [2]. Subsequently, research efforts sought to establish a conditional equivalence between PCNs and BP [9], with the goal of positioning PCNs as a biological instantiation of BP. It is important to clarify, however, that this equivalence is not general. The proof of equivalence relies on strict constraints regarding the network’s initialization, a precisely timed layer-wise parameter update schedule, and specific settings for the inference dynamics [9]. These requirements mean that PCNs align with BP only in a carefully engineered scenario, not universally. The paper under review [6], however, reverts to a more fundamental inquiry: it eschews the constraint of equivalence to BP to instead investigate the unique merits of this distinct algorithm. This constitutes a highly valuable research direction.

Key Computational Advantages of the PCN Framework

A key innovation of PCNs over classical BMs lies in their ability to bypass the requirement for “negative samples”. In BMs, learning is inherently contrastive: it demands not only lowering the energy of observed configurations (“positive samples”) but also raising the energy of unobserved, incorrect configurations (“negative samples”). This “negative phase” is computationally costly, as it effectively requires estimating the normalization term of the energy function—a primary obstacle to scalability. PCNs address this challenge by defining network energy as the sum of squared prediction errors, where the normalization term reduces to a constant. As a result, the learning process is noncontrastive: minimizing the energy of neural activities that achieve mutual prediction across layers automatically separates them from incompatible configurations, without explicitly sampling negatives. Recent work [10] has shown that exploiting this principle can substantially enhance the generative capacity of EBMs.

Beyond their advantage over classical EBMs, PCNs also offer distinctive benefits relative to BP. In their experiments [6], PCNs were compared with standard artificial neural networks trained via BP under equivalent architectures and loss functions, with key metrics including target alignment and test error across training iterations. Quantitative analyses revealed that PC in PCNs substantially reduced interlayer interference, leading to higher target alignment and faster learning, particularly in the batch-size-1 regime and in deeper architectures. Consequently, PCNs under PC outperform BP when interference is prominent—such as in deep or online learning settings—while BP remains competitive when interference is minimal (e.g., shallow or large-batch training). These findings position PCNs as a biologically grounded yet computationally efficient alternative learning paradigm to BP under specific regimes.

A Re-examination: Relationship with Backpropagation and Biological Plausibility

Nevertheless, assertions regarding the superior biological plausibility of PCNs compared to BP demand careful examination. The most frequently cited biological implausibilities of BP are 3-fold [11]: (a) the stipulation of symmetric weights for feedforward and feedback connections (b) the nature of error signals as signed real numbers, and (c) the principle that feedback signals do not influence neural activity during the forward pass.

The standard formulation of PCNs, in fact, also manifests the first 2 issues: it mandates symmetric weights (although some research indicates that asymmetric connections may be viable under specific constraints [12]), and its error signals are similarly signed. It is on the third point that PCNs present a distinct departure from BP. Within a PCN, top-down predictions (feedback) are integral to the inference process of guiding the system to an energy minimum; feedback and feedforward processes are thus inextricably linked. From this perspective, PCNs should be viewed not as a biologically superior alternative to BP but rather as a fundamentally different learning paradigm—one that formulates learning as the dynamics of inference within an energy-minimizing system.

Challenges and Future Directions

Notwithstanding their merits, PCNs confront substantial challenges that also define critical opportunities for future research.

Computational overhead: The inference process in a PCN is considerably more computationally demanding than that in a standard feedforward network trained with BP. BP entails a single forward pass to generate an output and a single backward pass to compute gradients. In marked contrast, the inference phase of a PCN necessitates an iterative relaxation process, wherein neural activities must dynamically evolve over time to locate a minimum in the energy landscape. This iterative search for an energy equilibrium can be computationally intensive, rendering PCNs slower for inference. This limitation, however, may be a reflection of a fundamental incongruity with contemporary von Neumann architectures, which segregate memory and computation. Neuromorphic, in-memory computing hardware could prove to be substantially better suited for this computational paradigm.

Architectural inflexibility: The efficacy of backpropagation stems from its generality. Provided that a computational graph of differentiable operations can be constructed, BP is capable of computing gradients via the chain rule. This has facilitated a "Cambrian explosion" of architectural diversity, including the remarkably successful transformer model. PCNs, by contrast, are more constrained by their specific, meticulously structured energy function. The adaptation of this framework to arbitrary neural computations—such as the multiplicative, second-order interaction (QK^T) central to the transformer's self-attention mechanism—is nontrivial. Such an adaptation would likely necessitate complex and difficult-to-stabilize higher-order energy terms.

The challenge of temporal modeling: PCNs face inherent difficulties in processing sequential data, as they lack a mechanism analogous to "backpropagation through time" (BPTT). Learning temporal dependencies requires that error signals propagate across time, whereas the energy minimization in a PCN is an instantaneous process. Although some research has endeavored to apply PCNs to sequential tasks [13], it has succeeded only in achieving a one-step approximation of BPTT. This limitation, however, may suggest that the brain employs different strategies for resolving temporal problems. One hypothesis is that the brain may avoid explicitly accumulating errors over time, as required by BPTT, by instead relying on structured and stable representations. Numerous experimental studies have provided converging evidence that the hippocampal–entorhinal system functions as a "cognitive map" [14,15]. Complementary computational theories have further interpreted this system from the perspective of cognitive mapping [16,17], highlighting its role in structuring spatial and temporal representations. Building on these insights, Dong et al. [18] directly leveraged the notion of cognitive maps to enhance the representational capacity of PCNs. In this framework, cognitive maps offer a stable representational scaffold that enables temporal prediction errors to converge toward consistent internal states. Consequently, a sequential task can be reformulated as a quasi-static inference problem—one that naturally aligns with PCN mechanisms. An alternative avenue involves the integration of PCNs with online learning algorithms [19]. Through the continuous and prospective adjustment of the network based on incoming information, PCNs might be adapted for a wider spectrum of temporal learning tasks, thus offering new perspectives on how the brain learns within a dynamic environment.

A forward-looking perspective: The challenges outlined above crystallize into several concrete and compelling research questions that could shape the future of predictive coding. First, in the realm of hardware and algorithms, how can we co-design neuromorphic architectures and PCN algorithms to fully exploit their natural synergy for efficient inference? Second, to overcome architectural rigidity, can hybrid models be developed that strategically combine fast, flexible BP-based modules with PCN-based modules that enforce global consistency and principled inference? Finally, to address temporal learning, beyond single-step approximations, what novel mechanisms, such as dynamic or hierarchical cognitive maps, can equip PCNs to effectively model and predict complex, long-range temporal sequences? Exploring these questions will not only advance PCNs as a computational framework but also deepen our understanding of the brain's remarkable efficiency and flexibility.

Conclusion: A Different Path to Intelligence

By furnishing a robust, biologically inspired framework for credit assignment, this paper makes a substantial contribution to the field. While the computational overhead and architectural constraints are likely to impede the widespread adoption of PCNs as a universal substitute for backpropagation in the immediate future, they conspicuously delineate a potent, alternative trajectory for the development of intelligent systems. The unique properties of PCNs may prove highly advantageous in specific domains and offer a rich theoretical foundation for comprehending the brain's own extraordinary learning capabilities.

Acknowledgments

Competing interests: The authors declare that they have no competing interests.

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